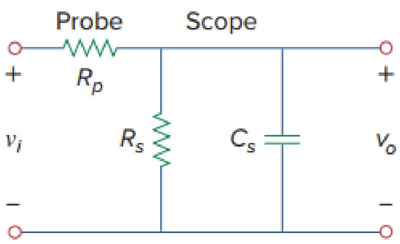
Chapter 7, Problem 90CP

Problem

An attenuator probe employed with oscilloscopes was designed to reduce the magnitude of the input voltage *vi* by a factor of 10. As shown in Fig. 7.149, the oscilloscope has internal resistance *Rs* and capacitance *Cs*, while the probe has an internal resistance *Rp*. If *Rp* is fixed at 6 MΩ, find *Rs* and *Cs* for the circuit to have a time constant of 15 μs.

Fig. 7.149:

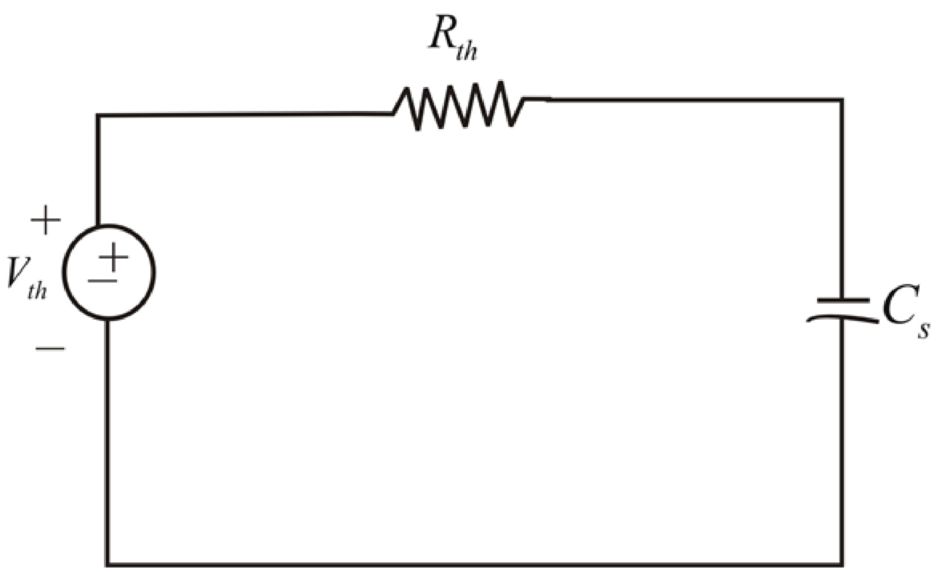


Step-by-step solution

Step 1 of 2

We determine the Thevenin's equivalent circuit for the capacitor CS.

$$V_{th} = \frac{R_S}{R_S + R_P} V_i \quad , \quad R_{th} = R_S \parallel R_P$$



Step 2 of 2

The Thevenin’s equivalent is an RC circuit, Since,

$$V_{th} = \frac{1}{10}V_i \rightarrow \frac{1}{10} = \frac{R_S}{R_S+R_P}$$

$$R_S = \frac{1}{9}R_P = \frac{6}{9} = \frac{2}{3}M\Omega$$

Also,

$$\tau = R_{th}C_S = 15\mu S$$

Where,

$$R_{th} = R_P \parallel R_S = \frac{6(2/3)}{6+2/3} = 0.6M\Omega$$

$$C_S = \frac{\tau}{R_{th}} = \frac{15\times10^{-6}}{0.6\times10^{+6}} = 25PF$$